SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – CASE STUDY

Course Code:	CSA0615	**Course Title:**	Design and Analysis of Algorithms
CO Assessed:	**CO3:** Articulation (BL4, BL5)	**Assessment:**	Assessment Tool 3 – Case Study
Weightage:	20% of CIA	**Total Marks:**	100 (2 Questions × 50 Marks)
CO3 Statement:	Design Divide and Conquer algorithms for Merge Sort, Quick Sort, Binary Search and Matrix Multiplication		
Student Name:	Soundariya M	**Reg. No.:**	192524342
Section / Batch:	Btech-AI&DS	**Date:**	20:08:2026

Instructions to Students

- Answer both questions. Each question carries 50 Marks. Total: 100 Marks.
- Carefully analyze the given case study questions.
- Explain in detail with sample code if needed.
- Clearly identifies all key issues, in-depth analysis, Innovative, feasible solutions with strong justification

Question Type	Description	Questions	Total Marks
Case Study	Focus on Design Divide and Conquer algorithms: Merge Sort, Quick Sort, Binary Search and Matrix Multiplication	Q1 – Q2	100

SECTION A – CASE STUDY

Question 1:

QUESTIONS

Q1
Divide and Conquer
50 marks

Q2
Divide and Conquer
50 marks

2/2 answered

Q1 **Divide and Conquer**

50 marks

Case Study: Search Engine Query Processing (Binary Search)

A search engine maintains a sorted index of keywords and must retrieve results quickly. Analyze how Binary Search improves search efficiency. Identify key requirements such as sorted data and fast retrieval. Apply divide and conquer principles to explain the process. Analyze time complexity and performance. Design a solution and justify why Binary Search is optimal in this context.

Your Code

```
def binary_search(arr, x):
    low = 0
    high = len(arr) - 1

    while low <= high:
        mid = (low + high) // 2

        if arr[mid] == x:
            return mid  # found index
        elif arr[mid] < x:
            low = mid + 1
        else:
            high = mid - 1

    return -1  # not found


n = int(input().strip())
arr = list(map(int, input().split()))
x = int(input().strip())

idx = binary_search(arr, x)

if idx != -1:
    print("Found at index:", idx)
else:
    print("Not Found")
```

Input

```
5
10 20 30 40 50
30
```

Output

```
Found at index: 2
```

Submitted

Question 2:

QUESTIONS

Q1
Divide and Conquer
50 marks

Q2
Divide and Conquer
50 marks

2/2 answered

Q2 **Divide and Conquer**

50 marks

Case Study: University Course Lookup System (Binary Search)

A university system maintains sorted course records and requires fast retrieval. Analyze how Binary Search improves search efficiency. Identify requirements such as sorted indexing and accuracy. Apply divide and conquer concepts. Analyze time complexity and justify the solution.

Your Code

```
# Binary Search - University Course Lookup System
# Easy input/output

def binary_search(arr, target):
    low, high = 0, len(arr) - 1

    while low <= high:
        mid = (low + high) // 2

        if arr[mid] == target:
            return mid  # found
        elif target < arr[mid]:
            high = mid - 1
        else:
            low = mid + 1

    return -1  # not found


# ---- Easy Input ----
n = int(input("Enter number of courses: "))

courses = input("Enter sorted course codes (space separated): ").split

target = input("Enter course code to search: ")

index = binary_search(courses, target)

if index != -1:
    print(f"Course {target} found at index {index}.")
else:
    print(f"Course {target} not found.")
```

Input

```
6
101 105 110 115 120 130
115
```

Output

```
Enter number of courses: Enter sorted course codes (space separated): Enter course code to search: Course 115 found at index 3.
```

Code of Conduct

*I **Soundariya M/192524342**(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: **Soundariya M**

RUBRICS FOR CALCULATION

Case Study					
Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Mark Evaluation:

Case Study	Understanding of case (25)	Applications of concepts (25)	Analysis (20)	Solution Design (20)	Clarity (10)	Total Marks

						(Each - 50)
Q1						
Q2						
					Overall Total (100)	

Faculty In-charge	Course Coordinator	Head of Department
Name: ___________________ Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________